

Adaptive-Assisted Imitation-to-TD Training for Top-Down Single-Block Lift on the Unitree G1 with Dex3 Hand

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Abstract—Contact-rich dexterous manipulation is difficult for model-free reinforcement learning because sparse task rewards are gated by millimeter-scale geometry and discontinuous contact events. The original project proposal hypothesized that a FastTD3-only reinforcement-learning pipeline could solve progressive humanoid block stacking [6]; in practice, that approach failed to reliably discover stable grasp-and-lift behavior and motivated the pivot studied here. We study a 6 cm single-block lift task on a simulated Unitree G1 humanoid with a Dex3 hand, using a top-down reach, two-finger pinch grasp, and adaptive-assisted imitation-to-temporal-difference (TD) training method. The method combines a scripted damped-least-squares inverse-kinematics (DLS-IK) contact teacher, weighted behavior cloning (BC), DAGger-style teacher relabeling of policy-visited states, replay-inclusive handoffs, privileged-critic observations, a stage-routed reward, and FastTD3. The central design is to treat the teacher not as a separate controller to be evaluated, but as a state-labeling and assistance mechanism that shapes the replay distribution while the deterministic actor gradually assumes control. Preliminary results are presented for 150k teacher-only steps, 500k noisy DAGger/BC steps, and a replay-resumed adaptive FastTD3 continuation after the teacher/DAGger hand-off. We formalize the curriculum MDP, teacher policy, action-gated behavior distribution, adaptive assist schedule, supervised loss, and FastTD3 objectives. The pipeline reaches contact and transient lift under high assist, while a pure-FastTD3 approach produced fingertip contact but failed to generate strict contact or measurable lift in the observed training window using the same observations and reward function.

Index Terms—reinforcement learning, temporal-difference learning, behavior cloning, DAGger, TD3, FastTD3, humanoid manipulation, Unitree G1, Isaac Lab

I. INTRODUCTION

FastTD3 shows that off-policy TD3-style learning can scale to high-dimensional humanoid-control tasks using parallel simulation, a distributional critic, and tuned reward design [1]. Related TD-based humanoid locomotion work reports simulated Unitree H1-2 results on high-dimensional contact-rich motions [2]. In the top-down lift task studied here, pure exploration from a random deterministic policy spends most of its budget outside the narrow contact manifold required for a stable grasp.

We address this exploration problem by combining imitation and off-policy TD learning. A scripted DLS-IK/contact teacher supplies low-variance geometric labels; a BC objective transfers these labels into the actor; DAGger-style relabeling

labels the states reached under assisted policy rollouts; and FastTD3 reuses the same replay distribution for critic and actor improvement. A reduced 14-dimensional action space keeps the policy surface small enough for continuous-control TD learning while retaining the seven right-arm joints and seven right-hand finger joints needed for the top-down grasp.

The contribution is a training formulation and a reproducible software pipeline for contact-rich manipulation. We formalize the teacher-assisted action distribution, observation and privileged-critic contracts, action-gated replay semantics, adaptive assist schedule, stage-routed reward, and FastTD3 update used by the system. The code base implements this as a single pipeline launcher with an internal stage-routed task: the full training sequence is declared in one Python entry point, and replay-bearing checkpoints are written for continuation. We also report preliminary single-seed training diagnostics for two conditions: an RL-only learner and the proposed adaptive-assisted teacher transfer. These preliminary diagnostics are used to characterize the current failure mode rather than to claim solved autonomous lift.

II. RELATED WORK

TD3 and FastTD3. TD3 addresses overestimation in deterministic actor-critic learning using clipped double critics, target-policy smoothing, and delayed actor updates [3]. FastTD3 keeps the deterministic-policy backbone but uses a distributional critic and high-throughput parallel simulation, showing strong humanoid-control performance [1]. We use the distributional FastTD3 variant for the main adaptive-assisted learner.

Imitation and DAGger. BC minimizes supervised error against expert actions but is brittle under covariate shift: a small policy error can move the rollout into states the expert dataset never covered. DAGger addresses this by labeling states visited by the learner and aggregating them into the training set [4]. The implementation here is a deterministic continuous-control variant: execution actions are blended between teacher and policy rather than sampled from a stochastic expert/learner mixture, but the key DAGger idea remains – teacher labels are aggregated on states visited by the learner-influenced behavior policy.

PPO and SAC alternatives. PPO is on-policy and updates using data collected by a recent behavior policy [10]. We did not evaluate PPO here because the central experiment depends on reusing teacher/Dagger replay across the handoff to value learning. SAC is off-policy but uses a stochastic maximum-entropy objective [11]. We also did not evaluate SAC; the transfer method studied here intentionally keeps a deterministic actor close to teacher-shaped contact actions while the critic learns from replay.

Dexterous manipulation and simulation. Prior dexterous-hand work demonstrated learned in-hand object reorientation on a physical Shadow Dexterous Hand [7]. Our study is narrower: it isolates a simulated top-down pinch-and-lift primitive before perception, placement, or hardware transfer. The environment uses Isaac Lab [9], the successor to Orbit [5], and Unitree’s public simulation environment resources [8]. Damped least-squares inverse kinematics follows the standard DLS idea to stabilize IK near singularities [12].

III. PROBLEM FORMULATION

A. Task

The task is top-down approach, grasp, two-finger pinch, and lift of a single red 6 cm cube by a fixed-base Unitree G1 humanoid with a Dex3 right hand. The policy commands a reduced 14-dimensional action: seven right-arm joints and seven right-hand finger joints. Legs, waist, left arm, and left hand remain clamped. Training uses a unified stage-routed reward for reach, alignment, contact, and lift. The stage state routes reward terms, action gates, teacher behavior, and diagnostics, but the training objective remains the same across teacher-only collection, noisy DAGger/BC, and FastTD3 refinement. Success is defined as a gated sustained 2 cm lift with opposed contact and drift/tilt constraints.

We model the simulator as an augmented Markov decision process with physical state s_t , curriculum stage $c_t \in \{0, 1, 2\}$, hold-counter vector h_t , persistent latch vector L_t , and teacher/controller memory ζ_t . The full simulator state is

$$\begin{aligned} x_t &= (s_t, c_t, h_t, L_t, \zeta_t), \\ o_t &= h(x_t), \quad p_t = h_{\text{priv}}(x_t), \end{aligned} \quad (1)$$

where o_t is the public actor observation and p_t is privileged critic information. The actor receives o_t only, while the critic can receive (o_t, p_t) . The transition is

$$x_{t+1} \sim P(\cdot \mid x_t, a_t), \quad (2)$$

where the curriculum components of x_t are deterministic functions of geometry gates and hold counters. The scope is intentionally narrower than full stacking: no perception milestone, no placement stage, no multi-object policy, and no sim-to-real deployment claim.

B. Training Architecture

The learning system has four coupled components. First, a geometry teacher computes arm and hand actions from block-relative pose. Second, observation-conditioned action gates and action-surface maps constrain live execution and

the actor/critic objectives. Third, a replay buffer stores policy-assisted transitions together with teacher relabels. Fourth, an asymmetric actor-critic learner uses public observations for the actor and privileged controller/teacher state for the critic. This design makes the data distribution, supervised labels, and TD targets consistent across teacher-assisted and lower-assist phases.

C. Observation and Action Contract

The actor observes a 171-dimensional public vector containing robot state, palm/fingertip geometry, shaping scalars, contact/lift summaries, and curriculum signals. The critic additionally receives privileged controller, teacher, IK, and latch state. Fig. 1 summarizes the actor and privileged channels. This asymmetric setup is appropriate for simulator learning but not directly deployable on hardware without replacing privileged signals by estimated perception/tactile channels.

Action gates are modeled as deterministic operators on proposed actions. Finger-closure commands are gated by thumb/index tip error to the block-centered contact targets, closure can be capped after contact, and middle-finger commands can mirror index commands. Live execution, replay storage, and TD target construction use related but separately configured action surfaces. Fig. 2 shows how the same geometry gates route stage rewards, teacher closure, diagnostics, and action masks. This matters because off-policy replay can contain actions collected under different assist and gate states. The implementation therefore separates the live execution gate from the configurable action surface used by BC and actor-critic objectives.

Let a proposed reduced action be split into arm and finger blocks, $u = [u_A, u_F]$. The gate operator is

$$G(o, u) = C_{\text{cap}} \left(o, M_{\text{mir}} \left[\begin{array}{c} u_A \\ a_{\text{floor}}(o) + g(o) \odot [u_F - a_{\text{floor}}(o)]_+ \end{array} \right] \right), \quad (3)$$

where $g(o) \in [0, 1]$ is a geometry-derived close/unlock progress scalar that is broadcast over the finger channels, $a_{\text{floor}}(o)$ is the pre-contact or preload finger floor, $[\cdot]_+$ clips negative closure above that floor, M_{mir} applies middle-finger mirroring when configured, and C_{cap} applies close caps and action bounds. The live gate maps proposed policy-level actions to executable environment commands. Replay stores the selected reduced action used for critic training, while target and objective actions are transformed through configured action maps. For update objectives, the trainer uses a configured action map

$$A_\eta(o, u) \in \{u, G(o, u), u + \text{stopgrad}(G(o, u) - u)\}, \quad (4)$$

corresponding to raw, environment-gated, and straight-through modes. The reported final runs use the raw objective surface for actor BC and actor-Q updates while still applying live gates before environment execution.

Actor observation: 171 public dimensions		
Group	Dims	Contents and role
Robot state	101	29 body joints as position/velocity/torque plus 14 Dex3 finger positions. Gives the actor proprioception, including held/unused joints.
Task geometry	38	Palm, block, grip target, thumb/index/middle tips, and fingertip-to-block vectors. This is the geometric state used for reach and contact alignment.
Shaping scalars	27	Palm pose errors, axis errors, alignment, contact strengths, lift height, tip-target errors, physical block scalars, and source-block one-hot.
Curriculum signals	5	Stage one-hot, normalized episode step, and final-column finger unlock/close progress used by action gates.
Privileged critic / teacher-side observation: 78 dimensions		
Group	Dims	Contents and role
Controller state	22	Curriculum latches, hold counters, stage-2 age, contact-pose readiness, finger-center readiness, lift latches, and arm-hold state.
Contact teacher state	34	Teacher FSM flags, closure fractions, contact latches, missing-contact flags, descent amounts, and tip/center servo magnitudes.
IK teacher state	14	DLS/top-down state: inward motion, contact alignment, planar alignment, pocket sweep, and lift-freeze state.
Arm-hold target	8	Frozen waist + right-arm hold target used by the critic to disambiguate teacher/controller state.
Public actor input $\rightarrow$ actor action; public input + privileged state $\rightarrow$ critic value; teacher/controller state $\rightarrow$ BC labels, assist blend, and DAgger replay labels.		

Fig. 1. Observation contract. The actor gets only the 171-dimensional public vector. The critic can additionally consume privileged controller and teacher state, making the simulator training problem easier but not directly hardware deployable.

D. Curriculum and Reward

The curriculum is a finite-state controller embedded in the MDP. For each stage k , a Boolean geometry predicate $\Phi_k(x_{t+1})$ is debounced by a stage-local hold counter,

$$h_{t+1}^{(k)} = \begin{cases} \min(H_k, h_t^{(k)} + 1), & c_t = k \wedge \Phi_k(x_{t+1}), \\ 0, & c_t = k \wedge \neg \Phi_k(x_{t+1}), \\ h_t^{(k)}, & c_t \neq k, \end{cases} \quad (5)$$

and the stage update is

$$c_{t+1} = \min\left(2, c_t + \mathbf{1}\left[h_{t+1}^{(c_t)} \geq H_{c_t}\right]\right). \quad (6)$$

Persistent contact and lift latches are monotone Boolean state variables within an episode,

$$L_{t+1} = (L_t \vee \Psi(x_{t+1})) \wedge \neg R_{t+1}, \quad (7)$$

where R_{t+1} denotes episode reset or controlled fallout.

With $y_t = (x_t, a_t, x_{t+1})$, the reward is a stage-routed transition reward:

$$r_t = \sum_{k=0}^2 \mathbf{1}[c_t = k] \sum_j w_{k,j} \phi_{k,j}(y_t) - \sum_m \lambda_m \rho_m(y_t). \quad (8)$$

Here $\phi_{k,j}$ are reach, alignment, contact, and lift terms, while $\rho_m \geq 0$ are global penalty magnitudes such as drift, tilt, angular velocity, uncentered lift, and drop with $\lambda_m \geq 0$. The canonical run fixes this full-reward vector across teacher-only, DAgger/BC, and RL/adaptive collection; the schedule changes behavior policy, supervised labels, assist, and optimizer settings rather than switching to separate external reward phases.

IV. METHOD

A. Teacher and Adaptive Assist

The teacher combines DLS-IK arm targets with scripted contact closure. Its output is a reduced action $a_t^* = [a_{A,t}^*, a_{F,t}^*]$, where the arm component tracks a top-down palm target and the finger component follows a bounded closure schedule driven by stage, contact, missing-tip, and lift-latch state.

The arm teacher solves a small Cartesian tracking problem at each step. Let $q_a \in \mathbb{R}^7$ be the right-arm joint vector, $x(q_a) \in SE(3)$ the palm pose, and x_d the top-down palm target generated from block pose and grip geometry. The pose residual is represented as

$$e = \begin{bmatrix} p_d - p(q_a) \\ \text{Log}(R_d R(q_a)^\top) \end{bmatrix} \in \mathbb{R}^6, \quad (9)$$

where the first three components are translational error and the last three are spatial-frame axis-angle orientation error. With geometric Jacobian $J(q_a) \in \mathbb{R}^{6 \times 7}$ in the same frame, damped least-squares IK computes

$$\Delta q_{ik} = J^\top (J J^\top + \lambda^2 I)^{-1} e. \quad (10)$$

For the redundant seventh arm degree of freedom, the teacher can bias the solution toward a nominal open-hand posture q_{nom} through a damped nullspace-like term,

$$\begin{aligned} J_\lambda^\# &= J^\top (J J^\top + \lambda^2 I)^{-1}, \\ N_\lambda &= I_7 - J_\lambda^\# J, \\ \Delta q_a &= \Delta q_{ik} + \alpha N_\lambda (q_{nom} - q_a). \end{aligned} \quad (11)$$

Because $J_\lambda^\#$ is damped, N_λ is a bias projector rather than an exact Moore–Penrose nullspace projector.

Reward routing			
Stage	Active reward family	Main signals	
Stage 0: reach	Reach and pre-alignment terms	Palm distance/height/orientation, yaw/spread axes, fingertip-line angle, open-hand alignment, reach shell bonus.	
Stage 1: align	Alignment and contact-preparation terms	Stricter palm geometry, opposed-face gate, fingertip-line angle, alignment error, no-contact bound, align shell bonus.	
Stage 2: contact/lift	Contact, grip, lift, and safety terms	Opposed contact, target distance, vertical gap, centered contact, lift progress, lift-with-grip, drift/tilt/drop penalties.	
Unified task reward	Stage-routed full reward	Single reward function combines the reach, align, and contact/lift stages with a gated 2 cm lift objective, opposed-contact requirement, and reduced but nonzero drift, tilt, drop, velocity, and uncentered-lift penalties.	
Curriculum geometry gates			
Palm geometry	Finger geometry	Contact geometry	State/action effect
Palm distance, height, orientation, yaw axis, spread/drop axis	Thumb/index/middle tip locations, fingertip-line angle, block-center XYZ residuals	Thumb/index contact, opposed contact, contact-pose shell, centered-contact readiness	Stage holds advance 0→1→2; finger unlock/XYZ-close gate opens; teacher closure fractions update; target and objective actions use configured action maps.
Geometry metrics → debounced hold counters/latches → stage mask $g(x)$ → reward terms + teacher FSM + action gates → replay rows and TD targets.			

Fig. 2. Reward and curriculum-gate flow. The stage machine and reward function are coupled through geometry-derived masks; the proposed pipeline keeps one unified stage-routed reward task while changing data collection, assist, and loss modes over time.

Contact-specific corrections use a bounded pocket sweep. The sweep evaluates a small set of one-coordinate perturbations around the IK action and keeps the candidate that improves thumb/index pocket error. Let

$$q_0^{\text{des}} = q_a + \Delta q_a \quad (12)$$

be the unswept desired arm joint target, let $e_\tau(q)$ and $e_\iota(q)$ be first-order thumb and index pocket target errors predicted at joint target q , and let $\mathcal{C}(q_0^{\text{des}})$ be the candidate set formed by bounded one-coordinate joint offsets after soft-limit and max-step clipping. The sweep selects

$$q_{\text{sweep}} = \arg \min_{q \in \mathcal{C}(q_0^{\text{des}})} (\|e_\tau(q)\|_2^2 + \|e_\iota(q)\|_2^2). \quad (13)$$

The final teacher action is the normalized and clipped reduced-action command

$$a_t^* = \text{clip}(S[q_{\text{teach}}, f_{\text{close}}(\zeta_t)], -1, 1), \quad (14)$$

where $q_{\text{teach}} = q_{\text{sweep}}$ when the sweep improves the pocket score and $q_{\text{teach}} = q_0^{\text{des}}$ otherwise. The map S converts desired arm joints and absolute finger targets into the 14-dimensional policy action surface. In the contact pocket, arm changes are attenuated and closure plus sweep corrections dominate, producing lower-variance BC labels.

The executed behavior policy blends the actor and teacher, then applies the action gate:

$$\mu_{\beta_t}(o_t) = G(o_t, (1 - \beta_t)\pi_\theta(o_t) + \beta_t a_t^*). \quad (15)$$

The scalar $\beta_t \in [\beta_{\min}, 1]$ controls teacher assistance. Earlier adaptive runs reduced β_t when actor-to-teacher errors

remained small over a hold window. With arm/finger index sets A and F ,

$$\begin{aligned} e_t^A &= |A|^{-1} \sum_{m \in A} (\pi_\theta(o_t)_m - a_{t,m}^*)^2, \\ e_t^F &= |F|^{-1} \sum_{m \in F} (G(o_t, \pi_\theta(o_t))_m - G(o_t, a_t^*)_m)^2. \end{aligned} \quad (16)$$

Let N_g and N_b count consecutive windows satisfying good or bad tracking thresholds. The error-gated adaptive update is

$$\beta_{t+1} = \begin{cases} \max(\beta_{\min}, \beta_t - \Delta_-), & N_g \geq H_g, \\ \min(\beta_0, \beta_t + \Delta_+), & N_b \geq H_b, \\ \beta_t, & \text{otherwise.} \end{cases} \quad (17)$$

This error-gated rule was useful for debugging BC, but it was too conservative for the final objective: low actor-to-teacher error mostly verifies imitation, not task improvement. The pipeline therefore uses a strict-contact metric gate after the fixed DAGger/BC block. For a completed episode e , define the strict-contact fraction

$$q_{\text{strict}}^{(e)} = \frac{1}{T_e} \sum_{t=1}^{T_e} \mathbf{1}[\text{strict contact}_{e,t} = 1]. \quad (18)$$

Let $q_{\max}$ be the maximum strict-contact fraction over the populated rolling window of completed training episodes. Assist is lowered if $q_{\max} \geq 0.05$ and raised if the populated window contains no such episode:

$$\beta_{t+1} = \begin{cases} \max(\beta_{\min}, \beta_t - 0.005), & q_{\max} \geq 0.05, \\ \min(1, \beta_t + 0.010), & q_{\max} < 0.05. \end{cases} \quad (19)$$

Before the metric window is populated, β_t is held fixed. This gate deliberately measures whether the train-environment behavior still reaches the contact manifold, rather than whether the actor exactly matches an imperfect teacher. At the RL switch, policy BC relabeling and teacher-BC loss weights are disabled; teacher actions may still be computed for assisted behavior until assist reaches the configured floor.

B. Behavior Cloning

BC trains the actor to match teacher labels on the configured actor objective surface used by the trainer:

$$\mathcal{L}_{\text{BC}} = \frac{1}{14B} \sum_{i=1}^B \sum_{m=1}^{14} \omega_{im} (A_{\eta_{\text{BC}}}(o_i, \pi_{\theta}(o_i))_m - a_{i,m}^*)^2. \quad (20)$$

The weights ω_{im} allow arm and finger channels to be emphasized differently. Depending on configuration, $A_{\eta_{\text{BC}}}$ can be the raw reduced action, the replay-observation-gated executable action, or a straight-through gated action on the actor side. In the reported final runs, BC is applied on the raw reduced action surface so the actor is trained toward the teacher's intended command even when live gates temporarily block premature finger closure. In pure BC this objective is the full actor update. During TD training it becomes a regularizer on teacher-labeled replay rows when enabled, so the actor objective can trade off value improvement against remaining close to the teacher in contact-sensitive coordinates. Because the teacher can depend on controller memory not visible in o_i , BC minimizes conditional risk under the public observation rather than guaranteeing exact teacher recovery.

C. DAgger-Style Replay Aggregation

During teacher-only and DAgger/BC collection, the finite replay buffer stores both environment experience and a teacher BC target:

$$\mathcal{B}_{k+1} = \text{ReplayUpdate}(\mathcal{B}_k, o_t, p_t, a_t, R_t, \Gamma_t, o_{t+n}, p_{t+n}, d_t, \tau_t, a_t^*, \iota_t), \quad (21)$$

where R_t is the stored accumulated return over the configured replay horizon, Γ_t is the corresponding stored discount, τ_t marks timeout rows, and ι_t marks whether a teacher label is available. During the assisted RL phase, teacher actions may still be computed for assist/provenance, but policy BC relabeling is disabled. The state x_t is sampled under the behavior policy μ_{β_k} . Thus the supervised distribution is induced by states visited under assisted policy rollouts, not only by states visited by the teacher alone:

$$\min_{\theta} \mathbb{E}_{(o, a^*) \sim \mathcal{B}} \left[\sum_m \omega_m(o) (A_{\eta_{\text{BC}}}(o, \pi_{\theta}(o))_m - a_m^*)^2 \right]. \quad (22)$$

The implementation is a deterministic continuous-control variant of DAgger rather than the literal stochastic schedule analyzed in [4]; the relevant design principle is dataset aggregation on states induced by μ_{β_k} . The no-regret bound from the original DAgger analysis is therefore not claimed for high-assist runs where the state distribution remains teacher-biased.

D. Temporal-Difference Learning

TD learning estimates return by bootstrapping from the critic rather than waiting for complete rollouts [13]. For deterministic policy π , the discounted return and value are

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}, \quad (23)$$

$$Q^{\pi}(o, p, a) = \mathbb{E}_{\pi}[G_t \mid o_t = o, p_t = p, a_t = a].$$

The critic is training-only and conditions on replay samples (o, p, a) ; the deployed actor remains $\pi_{\theta}(o)$ and never consumes p . The stored n -step TD error for a target network is

$$\delta_t = R_t + \Gamma_t Q_{\bar{\psi}}(o_{t+n}, p_{t+n}, \bar{a}_{t+n}) - Q_{\psi}(o_t, p_t, a_t), \quad (24)$$

where $\bar{a}_{t+n} = A_{\eta_Q}(o_{t+n}, \pi_{\bar{\theta}}(o_{t+n}))$ and Γ_t includes the terminal mask for true terminal rows but not time-limit truncations. The canonical RL phase uses three-step replay targets. The critic minimizes δ_t^2 over replay. Following the deterministic policy-gradient theorem [14], the actor update differentiates the critic with respect to the actor output:

$$\begin{aligned} \nabla_{\theta} J \approx & \frac{1}{B} \sum_i \frac{\partial \pi_{\theta}(o_i)}{\partial \theta} \frac{\partial A_{\eta_Q}(o_i, u)}{\partial u} \Big|_{u=\pi_{\theta}(o_i)} \\ & \times \nabla_a Q_{\psi}(o_i, p_i, a) \Big|_{a=A_{\eta_Q}(o_i, \pi_{\theta}(o_i))}. \end{aligned} \quad (25)$$

Hard gates and clamps use the implementation's configured raw, gated, or straight-through routing; locked channels therefore contribute either raw actor gradient, zero/masked gated gradient, or straight-through gradient depending on the selected objective mode.

E. TD3 and FastTD3

For scalar TD3, the target-action form is

$$\begin{aligned} \varepsilon & \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c), \\ u' & = \text{clip}(\pi'(o') + \varepsilon, a_{\min}, a_{\max}), \\ \tilde{a}' & = A_{\eta_Q}(o', u'), \\ y & = R + \Gamma \min_j Q'_j(o', p', \tilde{a}'). \end{aligned} \quad (26)$$

The FastTD3 backend replaces each scalar critic with probabilities on fixed atoms,

$$\begin{aligned} p_{\psi_j, k}(o, p, a) & = \Pr_{\psi_j}[Z(o, p, a) = z_k], \\ z_k & = V_{\min} + k \frac{V_{\max} - V_{\min}}{K - 1}, \end{aligned} \quad (27)$$

for $k = 0, \dots, K - 1$. The corrected final-run configuration uses $K = 201$, $V_{\min} = -100$, and $V_{\max} = 100$. This widened support is required because the reward contains positive shaping and success/contact bonuses as well as penalties; a non-positive support such as $[-5, 0]$ would clip useful positive TD targets to the top atom and large penalties to the bottom atom. The scalar value used for actor updates and critic selection is the mean $Q_{\psi_j}(o, p, a) = \sum_k p_{\psi_j, k}(o, p, a) z_k$. The

distributional target is the C51 projection. With selected target critic probabilities $p_{\bar{\psi}_{j^*,k}}(o', p', \tilde{a}')$,

$$\begin{aligned}\hat{z}_k &= \text{clip}(R + \Gamma z_k, V_{\min}, V_{\max}), \\ m &= \Pi_{\mathcal{Z}}\{(\hat{z}_k, p_{\bar{\psi}_{j^*,k}})\}_{k=0}^{K-1}, \\ j^* &= \arg \min_{j \in \{1,2\}} Q_{\bar{\psi}_j}(o', p', \tilde{a}').\end{aligned}\quad (28)$$

Each online critic minimizes cross-entropy $-\sum_k m_k \log p_{\bar{\psi}_j,k}(o, p, a)$. The actor objective combines the deterministic value objective with any active BC regularizer:

$$\begin{aligned}\mathcal{L}_\pi &= -\frac{1}{B} \sum_i Q_{\text{act}}(o_i, p_i, A_{\eta_Q}(o_i, \pi_\theta(o_i))) \\ &\quad + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}.\end{aligned}\quad (29)$$

Here Q_{act} is the configured actor critic reduction: critic 1 for scalar TD3-style updates, or the clipped/averaged twin-critic reduction used by the selected FastTD3 configuration.

V. EXPERIMENTAL PROTOCOL

The diagnostic comparison asks whether teacher-shaped replay and adaptive assist move training into the contact regime faster than pure TD learning. Both conditions use the same robot, observation/action spaces, unified stage-routed reward weights, network class, simulator settings, and logged contact/lift diagnostics. The canonical teacher-transfer run uses one unified stage-routed reward task across teacher-only collection, DAGger/BC, and adaptive FastTD3; the pure-RL baseline removes teacher labels, BC, DAGger, and policy assist.

A. Proposed Pipeline

The final executable artifact is `training/curriculum/pipeline.py`, launched through `scripts/launch_pipeline.py`. It exists to remove ambiguity from the final experiments: all reward weights, success gates, optimizer settings, assist settings, replay settings, TensorBoard settings, and checkpoint settings are declared in one Python file before the run plan is materialized. The default run uses the unified stage-routed lift reward, including positive opposed-contact, bilateral-contact, centered-contact, lift-progress, lift-with-grip, and sustained-lift bonuses, plus reduced but nonzero drift/tilt/drop penalties.

The Proposed pipeline schedule is:

- 1) **Teacher-only replay collection:** 150k environment rows under $\beta = 1.0$, with assist noise disabled and no actor/critic updates. This seeds replay with clean top-down teacher trajectories.
- 2) **Noisy DAGger/BC:** 500k additional rows under assisted behavior. Execution actions receive arm/finger assist noise, but BC targets remain clean teacher relabels. The actor uses weighted BC on the raw reduced action surface with arm/finger weights 10 and 4.
- 3) **Replay-preserving FastTD3:** at 650k rows, the same replay buffer is retained, critic optimizers are reset, target networks are synchronized, training switches from

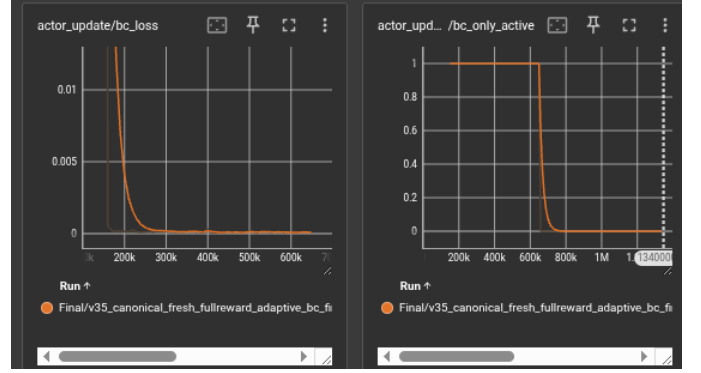


Fig. 3. Teacher/DAGger/BC handoff diagnostics. The first 150k rows are teacher-only replay collection with no actor update; BC losses and teacher-relabel regularization are active during the subsequent noisy DAGger/BC replay-aggregation window, then live relabeling/BC regularization is disabled at the 650k-row FastTD3 switch so the critic-driven actor can improve beyond teacher imitation.

BC-only updates to the FastTD3 RL-phase objective and hyperparameters, and policy BC relabeling is disabled.

- 4) **Strict-contact adaptive assist:** after the switch, β starts at 1.0 and is lowered by 0.005 only when the completed-episode metric window includes at least one episode with at least 5% strict contact. If the rolling completed-episode window loses that contact signal, β is raised by 0.010. The intended effect is to reduce teacher support only while the behavior policy remains on the contact manifold.
- 5) **Replay checkpoints:** checkpoints are written every 250k rows with replay included. A continuation uses `-resume-from`; replay and global-step resume are enabled by default and can be disabled with `-no-resume-replay` and `-no-resume-global-step`. The `-immediate-adaptive` option starts directly in the RL/adaptive phase when the checkpoint already contains teacher and DAGger replay.

Replay-inclusive continuation is supported by `-resume-from`. For continuation runs, the manifest records replay/global-step resume settings, the RL switch settings, the strict-contact gate threshold, and checkpoint saving flags. This makes it possible to distinguish a fresh run from a continuation of accumulated teacher/DAGger replay.

For evaluation, the behavior policy is disabled and the actor is run policy-only unless explicitly labeled assisted. The primary metric is gated held lift success,

$$S_{\text{hold}}^{(e)} = \mathbf{1} \left[\max_{0 \leq t \leq T-H} \sum_{\tau=t}^{t+H-1} \ell_\tau = H \right], \quad (30)$$

where $H = 60$ environment steps and

$$\ell_t = \mathbf{1} [z_t^{\text{block}} - z_t^{\text{table}} \geq 0.02 \wedge c_t^{\text{opp}} \geq 0.30 \wedge d_t^{xy} \leq 0.08 \wedge \theta_t^{\text{tilt}} \leq 45^\circ]. \quad (31)$$

Secondary metrics are single-step lift-above-threshold rate, maximum lift height, median best lift per episode, Stage-2 rate,

TABLE I
DIAGNOSTIC COMPARISON CONDITIONS.

Condition	Behavior policy	Replay labels / loss
RL only	$a_t = G(o_t, \text{clip}(\pi_\theta(o_t) + \epsilon_t)), \beta_t = 0$	No teacher labels; $\lambda_{BC} = 0$; TD/FastTD3 only.
Adaptive teacher	$a_t = \mu_{\beta_t}(o_t)$ with adaptive $\beta_t > 0$	Teacher relabels and weighted BC during DAgger/BC; replay retained for FastTD3 after 650k with RL-phase BC relabeling disabled.

opposed-contact rate, top-down success rate, actor-to-teacher arm/finger error, return, and assist coefficient over environment steps. Fixing these metrics before the final comparison is important because assisted rollouts can show contact or transient lift without proving that the actor has learned autonomous lift.

VI. PRELIMINARY RESULTS AND ANALYSIS

The first useful comparison is not final task success, but whether the proposed teacher $\rightarrow$ DAgger/BC front end moves the learner into the contact regime faster than pure RL under the same unified stage-routed reward. Fig. 4 compares training diagnostics from the teacher/Dagger/BC run against the pure-RL baseline: strict opposed contact, any fingertip contact, and mean lift height. These curves are not held-out policy-only evaluations, but they expose the central sample-efficiency difference that motivated the pipeline pivot. In this plotted single-seed diagnostic window, pure RL discovers occasional fingertip contact but shows little to no strict-contact signal and near-zero mean lift. The teacher/Dagger/BC front end was adopted to seed replay with states on the strict-contact and early-lift manifold before asking TD learning to refine the policy.

The submitted run sequence is still a training artifact rather than a completed success-rate result: it produces replay-bearing checkpoints from the teacher/Dagger schedule that continue into the strict-contact adaptive-assist FastTD3 phase. Fig. 5 shows the final continuation used for this handoff analysis. The run resumes from a replay-bearing checkpoint containing the teacher/Dagger handoff and early RL replay, then lowers policy assist only when completed train episodes retain at least 5% strict contact. Strict contact appears in bursts during the RL phase, including intervals that exceed the decay threshold, indicating that assisted training rollouts intermittently re-enter the opposed-contact regime. However, the signal is intermittent rather than steadily increasing; long stretches return near zero strict contact, so the actor has not learned a robust autonomous contact strategy.

These diagnostics identify the remaining bottleneck. The teacher and assist policy can drive the hand into the contact basin, but reaching strict contact intermittently is not enough: the actor must also learn a stable opposed-contact grasp that maintains height for the hold window. This is why the adaptive

FastTD3 phase gates assist decay on strict-contact episodes rather than actor-to-teacher BC error. The expected signature of success is therefore not a single transient lift or contact burst, but simultaneous improvement in held lift, opposed-contact rate, strict-contact rate, and lower-assist policy-only evaluation.

A. Current Bottleneck: Teacher-Bounded Transfer

The main unresolved issue is that the teacher is useful but not optimal. We ran an extended adaptive-assist FastTD3 phase whose practical goal was to reduce policy assist to the configured floor, $\beta_{\min} = 0$, while preserving the replay distribution seeded by teacher-only and DAgger/BC collection. This did reduce the problem to a cleaner transfer question, but it did not produce evidence that the autonomous actor improved beyond the pure teacher-shaped behavior. Available policy-only checks did not show conversion from contact into sustained held lift.

This failure mode is informative. The original adaptive rule based on actor-to-teacher error mostly measures imitation quality. If the teacher reaches contact with imperfect finger placement, one-sided contact, marginal squeeze, or occasional transient lift, then matching the teacher more closely can preserve those defects rather than remove them. The reward function is intended to provide the improvement signal, but in this task the reward surface is also highly shaped and gate-dependent: pose, contact, strict-contact, lift-progress, drift, tilt, and drop terms can disagree over short windows. Under high teacher assist, replay is concentrated near teacher-generated contact states; when assist is removed, small policy errors move the actor off that narrow manifold before the TD signal has demonstrated a robust correction.

The current interpretation is therefore conservative. The diagnostics show that teacher-shaped replay and action gates can move training into the contact regime, and that replay-inclusive continuation can preserve the teacher/Dagger distribution for TD learning. They do not yet demonstrate that the RL phase improves the policy beyond the teacher. The next technical step is to make the improvement target less teacher-bounded: evaluate assist decay using policy-only held-lift metrics, strengthen reward terms that distinguish stable opposed lift from teacher contact, and use the teacher primarily to seed replay rather than to define the actor’s long-term optimum.

VII. REPRODUCIBILITY

Each experiment records a manifest before training. The manifest includes the resolved configuration, trainer command, selected runtime environment, source hashes, input checkpoint hashes when present, run directory, and git state. This makes the training condition auditable without relying on ambient shell state or informal notes.

Replay provenance is recorded separately from model initialization. The training stack distinguishes actor-only initialization, full checkpoint resume, replay-inclusive resume, observation-stat transfer, frozen-policy teachers, and final

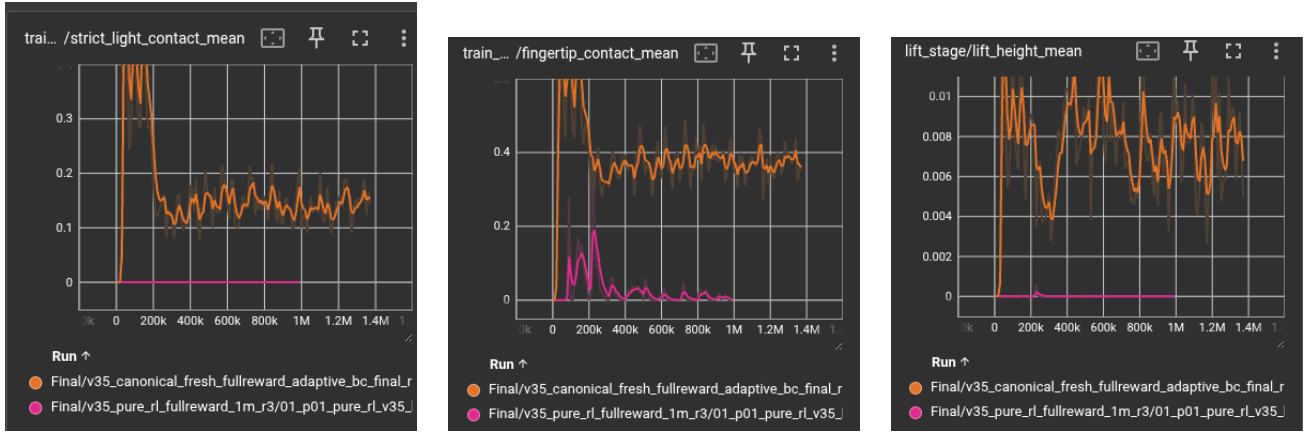


Fig. 4. Teacher $\rightarrow$ DAGger/BC transfer versus pure RL under the same unified stage-routed reward. Left: strict-contact diagnostic. Center: fingertip-contact diagnostic. Right: mean lift-height diagnostic. Pure RL produces some fingertip contact but little to no strict-contact signal and near-zero mean lift in the plotted window; the teacher-shaped run enters the strict-contact and transient-lift regime earlier. These are train-environment diagnostics, not final autonomous success rates.

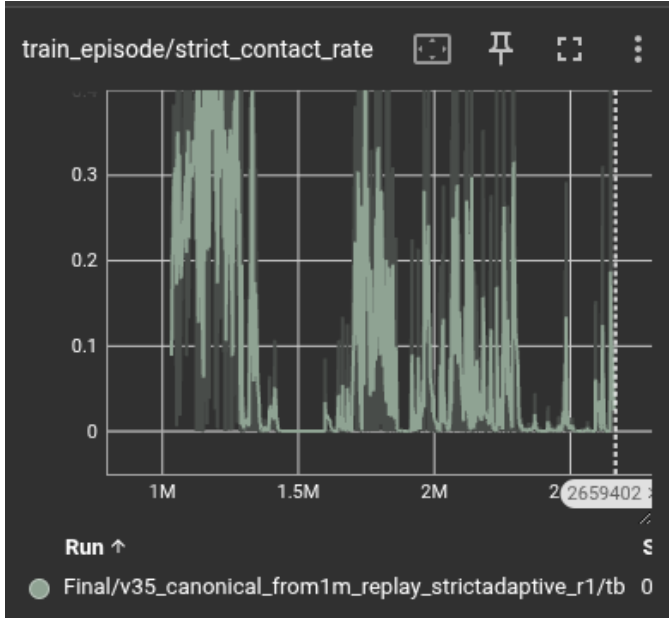


Fig. 5. Strict-contact-gated adaptive-assist FastTD3 phase. The plotted training diagnostic is the completed-episode strict-contact rate for the replay-resumed RL continuation. Contact bursts above the 5% decay threshold provide the task signal used to lower teacher assist, but the intermittent trace shows that strict contact is not yet stable.

handoff checkpoints. This prevents a common ambiguity in off-policy RL reporting: whether a continuation is learning from fresh experience or from the same accumulated replay distribution. In the proposed pipeline, replay-bearing checkpoints are mandatory because the central claim is a replay-shaped imitation-to-TD pipeline; a checkpoint is therefore meaningful only together with its manifest, replay-resume flags, and source hash.

VIII. LIMITATIONS

The study isolates top-down single-block lift rather than full block stacking [6]. It does not include camera perception, placement, stack construction, a multi-object policy, or hardware deployment. The actor still relies on simulator-derived geometric observations and action gates; the critic uses privileged state unavailable on hardware. The reported diagnostics are single-seed and should not be interpreted with population-level confidence.

The most important experimental limitation is that assisted rollouts and policy-only evaluations measure different objects. Assisted transfer can demonstrate that the teacher, gates, and replay distribution can reach contact, but it cannot prove autonomous lift. Matched policy-only evaluations for both RL-only and adaptive-assisted training are required, together with the assist trajectory used during data collection. A second limitation is teacher boundedness: when BC and adaptive assist are dominated by teacher agreement, the actor may inherit the teacher’s marginal contact strategy instead of discovering a more robust lift strategy from reward.

IX. CONCLUSION

This paper presented an adaptive-assisted imitation-to-TD method for top-down single-block lift on a Unitree G1 with a Dex3 hand. The method formalizes a DLS-IK/contact teacher, observation-conditioned action gates, DAGger-style replay re-labeling, adaptive teacher assist, weighted BC regularization, privileged-critic TD learning, and FastTD3 distributional critic updates under a single reduced action space. The submitted artifact makes the full training sequence explicit: 150k teacher-only replay rows, 500k noisy DAGger/BC rows with clean re-labels, a replay-preserving FastTD3 RL-phase switch, and strict-contact-gated assist decay with replay-inclusive checkpoints. The main empirical result is negative: the pipeline improves access to the contact regime and produces transient lift under high assist, but sustained autonomous held lift has not yet been demonstrated.

ACKNOWLEDGMENT

The author thanks Dr. Jerry Wu and the maintainers of Isaac Lab, FastTD3, and Unitree Robotics for open-source software and assets used in this project.

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